

Actions Homework

Overview

Your task is implement an `Action` that allows `Entities` to move. Movement code will be shared by all Entities in the game, so we must write it in a general way that can be used by us (humans) and monsters alike.

Steps

1. Create files `move.h` and `move.cpp` in folder `content/actions/`. Update the `CMakeLists.txt` in that folder.
2. Create class `Move` by inheriting from `Action`, overriding `perform()`. The `Move` class should take in the direction of movement in the constructor.
3. In `Move::perform()`, get the tile that the entity wishes to move to, inspect if you can move there. If you can then call `move_to()` on the Entity object and return `success()`, otherwise return `failure()`. Regardless make sure you always have the character facing the correct direction via `Entity::change_direction()`.
4. Bind movement directions to the arrow keys or WASD or both by adding a `behavior` function in `content/entities/heroes.h{cpp}`. You can determine the last key pressed using `engine.input.get_last_keypress()`.
5. Finally, bind the `Rest` action to any key.

Now your character should be able to move around with a room, but cannot yet open doors.

Testing

Ensure that the following happen:

- Walking towards an open tile, moves the character to that tile.
- Walking towards a wall, keeps the character on the same tile.
- Any attempts to move makes the entity face that direction.
- You are blocked by attempt to walk through doors.